	Topic to be covered	Unit #
1	Introduction	Unit 1
2	Alphabets, Strings, Languages	Unit 1
3	Finite Automata and its types, Regular Languages	Unit 1
4	Deterministic Finite Automata (DFA)	Unit 1
5	Problems on DFA construction	Unit 1
6	Non-deterministic Finite Automata (NFA)	Unit 1
7	Problems on NFA construction	Unit 1
8	NFA to DFA Conversion Algorithm	Unit 1
9	Problems on NFA to DFA Conversion	Unit 1
10	NFA with epsilon transitions, Equivalence of NFA and DFA	Unit 1
11	Thompson's Construction to Convert Regular Expression to NDFA, Subset Algorithm to convert NDFA to DFA	Unit 1
12	Minimization of DFA	Unit 1
13	Problems on Minimization of DFA	Unit 1
14	Moore machine construction techniques and problems	Unit 1
15	Mealy machine construction techniques and problems	Unit 1
16	Conversion of Moore machine to Mealy Machine & Vice-Versa	Unit 1
17	Families of Automata theory	Unit 1
18	Introduction to regular expression (RE)	Unit 1
19	Regular Expression to Finite Automata (FA) and vice-versa	Unit 1
20	Regular Grammar, Regular languages: Closure properties, product construction	Unit 1
21	Limitations of Automata Non-regularity: Pumping Lemma and Myhill–Nerode theorem	Unit 1
22	Context Free Grammar	Unit 2
23	Derivation tree and Ambiguity, Removing ambiguity, Parse Tress	Unit 2
24	Application of Context free Grammars, Closure Properties of CFG	Unit 2
25	Decidable of CFG	Unit 2
26	Push Down Automata: definition, the Languages of a PDA	Unit 2
27	Equivalence of PDAs and CFGs, Equivalence of PDAs accepting by final state and by empty stack,	Unit 2
28	Non-determinism in PDAs, Non-equivalence of NPDAs and DPDAs	Unit 2
29	Pumping Lemma for Context-Free Languages	Unit 2
30	Normal Forms for CFG's: Chomsky Normal Form and Greibach Normal form	Unit 2
31	Context–Sensitive Grammars, Linear Bounded Automata.	Unit 2
32	Universal Turing Machine, Post Machine, Deterministic Turing machines, Multi-tape TMs and its equivalence to single-tape TMs	Unit 3
33	Nondeterministic TMs and its equivalence to deterministic TMs	Unit 3
34	Chomsky Hierarchy, Post correspondence problem, Halting Problem	Unit 3
35	Turing enumerability, Turing machines Acceptability, Decidabilities, and Reducibility	Unit 3
36	Unsolvable problems about Turing machines	Unit 3
37	Time Complexity, P and NP complexity class	Unit 4
38	NP-complete problems (vertex cover, independent set and clique problem)	Unit 4